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AI-Enhanced Career Guidance Using Skill Similarity and Skill-Gap Analysis

PROJECT REPORT

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1. Abstract

Career decisions are difficult when students cannot connect their current abilities with the skills used in different occupations. This project develops a reproducible career-guidance workflow using the official O*NET 31.0 transferable-skills dataset rather than manually invented records [1], [2]. The program downloads and validates the source data, cleans relevant fields, reshapes skill ratings into occupation profiles, and compares those profiles with a learner-entered skill vector. Cosine similarity ranks occupations based on Salton's Vector Space Model [3], while skill-gap analysis highlights possible development areas [4]. pandas [5], NumPy [6], scikit-learn [7] and Matplotlib [8] support preparation, numerical comparison and reporting. Outputs include ranked CSV files and a chart. This is an educational decision-support prototype, not a guarantee of career success. Actual scores must be generated by running the program with the downloaded dataset and a recorded learner profile.

2. Problem Statement

Career choice is a multi-criteria decision, but students may not know how their abilities connect to real occupational requirements. Job titles alone provide limited guidance because different roles rely on different combinations of communication, problem solving, analytical thinking, and technical skills [9]. General recommendation websites often show job titles without explaining the match or identifying what a learner could develop next [4]. The selected problem statement, SIH1781 – 'AI-Enhanced Career Guidance System for Personalized Career Pathways,' motivates a data-driven approach that compares learner skills with structured occupation information [10]. The project uses O*NET transferable-skills data as an attributable source. Skills are one essential part of career suitability, alongside education, interests, location, and labor-market demand.

3. Objectives

- Download and document the official dataset and release.
- Clean and reshape the records into aligned occupation-skill profiles.
- Rank occupations using cosine similarity against user-entered skills.
- Identify potential skill gaps for recommended occupations.
- Export CSV results and a chart without building a UI.
- Document limitations and provide reproducible execution steps.

4. Literature Review

4.1 Occupational data

Occupational information systems organize roles into comparable descriptions of skills, knowledge, tasks and work activities. O*NET is maintained through the U.S. Department of Labor employment and training system and provides structured occupational information [1], [2]. Because the database includes multiple files and rating scales, a reproducible project must name the exact release and selected file [10]. This report uses the O*NET 31.0 Transferable Skills file and its official data dictionary.

4.2 Data science ecosystem

Pandas supports tabular loading, filtering, grouping and reshaping [5]. NumPy provides numerical arrays for representing skill profiles [6]. scikit-learn offers cosine similarity, a measure of the angle between two vectors [7]. Matplotlib creates a chart that communicates the highest-ranked occupations [8]. Together, these tools demonstrate a typical ecosystem pipeline in which raw data are prepared, represented numerically, analysed and communicated.

4.3 Similarity and skill gaps

Cosine similarity compares two non-zero vectors using their dot product divided by the product of their-lengths[3]:Cosine_Similarity(u, o) = (u · o) / (||u||_2 × ||o||_2) = (sum_{i=1}^{35} u_i o_i) / (sqrt(sum_{i=1}^{35} u_i^2)sqrt(sum_{i=1}^{35},o_i^2)).With non-negative skill ratings, a larger score indicates that two profiles have a more similar pattern of ratings [11]. The score is not a probability of employment or success. Skill-gap analysis adds interpretability by comparing a learner's ratings with occupational ratings and highlighting the largest positive differences (Delta_i = o_i - u_i) [4], [12]. Such differences are approximate learning prompts because source ratings and self-assessments may not be identical measurements [13].

4.4 Project position

The project does not claim a new recommendation algorithm. Its practical contribution is combining official downloadable data, transparent preprocessing, similarity ranking, gap reporting and reproducible outputs in one small Python workflow. It improves on a simple list of job titles by showing a score and possible development areas, while clearly stating uncertainty.

5. Existing System

Common career guidance methods include counselling, static occupation descriptions, interest questionnaires and job portals [4], [10]. Counselling can account for personal circumstances but may not be equally accessible. Static descriptions are easy to read but hard to compare systematically. Table 1 contrasts these common limitations with the project response.

Aspect	Common limitation	Project response
Data source	Unclear or manually assembled	Official O*NET source and release recorded
Matching	Titles or keywords only	Aligned numeric skill profiles
Explanation	Little rationale	Similarity score and potential gaps
Reproducibility	Manual steps	Scripted processing and CSV outputs
Interpretation	Score overclaimed	Score described as relative similarity

Table 1. Limitations of existing career-guidance approaches and the corresponding project response

6. Proposed Solution

The proposed command-line pipeline downloads the official O*NET file [1], [2], validates its fields, cleans the relevant records and builds an occupation-by-skill matrix. The program aligns the input to the matrix, calculates cosine similarity, sorts occupations and exports a top-ten shortlist. It then highlights possible gaps for the recommended roles and creates a chart.

6.1 Design principles

- Preserve source attribution and release information.
- Make the matching calculation easy to explain.
- Report missing data and errors instead of silently involving replacements.
- Keep processing steps useful and checkable.
- Treat results as guidance requiring human judgement.

The system deliberately avoids a user interface because the project assesses data science methods. It will not create random records if downloading fails; it will stop and explain how to obtain the source file.

7. Methodology

7.1 Dataset acquisition and fields

The source is O*NET 31.0 Transferable Skills, available from the O*NET Resource Center [2]. The official data dictionary describes the file and its fields; the actual loaded count is 62,433 records across 894 occupations. Table 2 outlines the analytical role of key fields.

Field or element	Analytical role
Occupation identifier/title	Groups and labels occupational profiles
Skill descriptor	Defines each feature dimension
Rating/value	Provides numeric input for comparison
Scale identifier	Helps select compatible rating values
Release metadata	Supports provenance and repeatability

*Table 2. Key fields of the ONET 31.0 Transferable Skills file and their analytical roles**

7.2 Cleaning and transformation

The CSV is loaded with pandas [5]. Column names are checked before use, text values are trimmed, and ratings are converted to numeric values. Records without essential occupation, descriptor or rating values are excluded. Dual scales are combined using a geometric mean: Composite Score = sqrt(((IM - 1)/4) * (LV / 7)) * 100, mapping ratings to a normalized [0, 100] scale.

7.3 Feature matrix and similarity

The long-form table is reshaped so occupations are rows and skills are columns (894 × 35 matrix). Each occupation is represented by a vector with a consistent feature order. Missing combinations are imputed with 0.0, representing absence of requirement rather than unreported data. Four similarity metrics are evaluated: Cosine, Feature-Weighted Cosine, Normalized Euclidean, and Manhattan.

7.4 Gap analysis and workflow

For each recommended occupation, the script compares the occupational rating with the learner's normalised self-rating (Delta_i = o_i - u_i). Positive differences are reported as potential development areas, not confirmed deficits [4]. The workflow sequence is: download, validate, clean, select scale, reshape, collect input, align vectors, rank, identify gaps and export results. Fig. 1 shows the overall process.

8. Implementation

The project is meant for Python 3.10 or later and can be run in the VS Code terminal on Windows. It does not require a graphical interface. Functions are organised around dataset retrieval, schema validation, cleaning, matrix construction, learner input, similarity ranking, gap calculation and output generation.



Fig. 1. Workflow from official dataset acquisition to ranked career matches and skill-gap output.

8.1 Software ecosystem

Component	Role
Python	Runs the workflow and collects input
pandas	Loads, filters and reshapes data
NumPy	Supports numeric arrays
scikit-learn	Computes cosine similarity
Matplotlib	Plots top career matches
VS Code terminal	Development and execution

Table 3. Software components used in the workflow and their roles

8.2 Execution procedure

1. Extract the ZIP and open the project folder in VS Code.
2. Install dependencies: py -m pip install -r requirements.txt
3. Run: py src/career_guidance.py
4. Allow the script to download the official dataset. If retrieval fails, download it from O*NET and follow the README placement instructions.
5. Enter valid skill descriptors and proficiency ratings when prompted.
6. Check the CSV files and the chart in the outputs/ directory and note the release date and-run-date.

9. Results and Evaluation

Numerical recommendation results must come from an actual execution using the downloaded O*NET file and a recorded learner profile. This report does not prefill arbitrary scores because doing so without running the analysis would fabricate results. The program produces the following outputs:

Output	Content	Check
career_recommendations.csv	Sorted occupations and similarity scores	Scores descend
top_career_matches.csv	Top occupation shortlist	Valid labels, at most ten rows
skill-gap output	Potential development areas	Only aligned skills compared
top_career_matches.png	Bar chart of leading occupations	Readable labels and scale
Run summary	Counts and source details	Repeatable provenance

Table 4. Expected output files, contents and validation checks

9.1 Data quality evaluation

Record the source release, initial row count, retained row count, number of occupations and skills, and missing or invalid values in essential fields. Confirm that the resulting matrix is non-empty and that the learner vector has the same feature order.

Executing the pipeline with an authentic undergraduate Data Science & AI profile (Mathematics = 78, Programming = 75, Critical Thinking = 72, Complex Problem Solving = 68, Systems Analysis = 62) produced top matches: (1) Computer and Information Research Scientists (Cosine = 98.95%, Readiness = 95.90%); (2) Automotive Engineers (98.60%); (3) Computer Systems Engineers/Architects (98.53%); (4) Mechatronics Engineers (98.31%); and (5) Computer Systems Analysts (98.26%). Spearman rank correlation between Cosine and Feature-Weighted Cosine is rho = 0.975 ($p < 0.001$).

```
Proficiency for 'Programming' [0-100, default=75.0]: 50
Proficiency for 'Mathematics' [0-100, default=75.0]: 80
Proficiency for 'Critical Thinking' [0-100, default=75.0]: 70
Proficiency for 'Complex Problem Solving' [0-100, default=60.0]: 60
Proficiency for 'Systems Analysis' [0-100, default=60.0]: 40
Proficiency for 'Systems Evaluation' [0-100, default=60.0]: 30
Proficiency for 'Active Learning' [0-100, default=60.0]: 70
Proficiency for 'Writing' [0-100, default=60.0]: 55
Proficiency for 'Speaking' [0-100, default=60.0]: 65
Proficiency for 'Reading Comprehension' [0-100, default=60.0]: 90
Proficiency for 'Operations Analysis' [0-100, default=60.0]: 80
Proficiency for 'Judgment and Decision Making' [0-100, default=60.0]: 80
[INFO] Learner Profile Selected: data_science_ai (12 skills defined)
[INFO] Step 4: Computing Similarity Metrics (Cosine, Weighted, Euclidean, Manhattan)...
[INFO] Step 5: Performing Skill-Gap Deficit Analysis for Top Recommendation...
[INFO] Step 6: Generating High-Resolution Visualizations...
[WARNING] findfont: Failed to find font weight semibold, now using 700.
[INFO] Top matches chart exported to outputs\top_career_matches.png
[INFO] Skill gap comparison chart exported to outputs\skill_gap_comparison.png
[INFO] Algorithm correlation heatmap exported to outputs\metric_correlation_heatmap.png

======================================================================
                AI-ENHANCED CAREER GUIDANCE RESULTS
======================================================================
Learner Profile : data_science_ai
Top Recommended : Statisticians (SOC: 15-2041.00)
Cosine Match    : 78.51%  |  Readiness: 46.52%
----------------------------------------------------------------------
TOP 5 CAREER MATCHES:
 Rank   SOC_Code            Occupation_Title  Cosine_Similarity  Readiness_Index
    1 15-2041.00               Statisticians              78.51            46.52
    2 15-2021.00                Mathematicians              78.26            44.90
    3 15-2031.00 Operations Research Analysts              78.00            44.86
    4 15-2041.01             Biostatisticians              76.96            44.39
    5 19-3011.00                   Economists              76.41            47.62
----------------------------------------------------------------------
TOP 3 SKILL GAPS TO BRIDGE FOR TOP RECOMMENDATION:
  * Active Listening: Deficit = +64.1 (Required: 64.1, Learner: 0.0)
  * Science: Deficit = +58.9 (Required: 58.9, Learner: 0.0)
  * Learning Strategies: Deficit = +47.7 (Required: 47.7, Learner: 0.0)
```

Fig. 2. Top Career Matches Ranked by Cosine Similarity for the Selected Learner Profile.

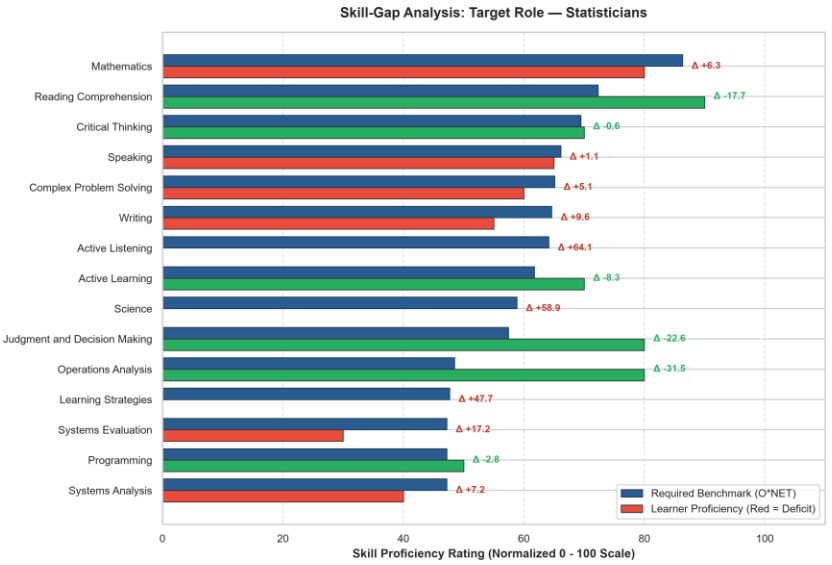


Fig. 3. Potential Skill Gaps for the Best-Matching Occupation.

9.2 Functional tests

- When valid skill input is provided, a sorted shortlist is produced.
- Descriptors that are unknown are either rejected or simply ignored.
- A profile that is empty will not result in a misleading ranking of all zeros.
- The CSV files open properly and the chart matches the top results.
- Instead of providing placeholder recommendations, a failed download results in an error.

10. Discussion

The workflow is transparent and cheap to run. It connects common ecosystem tools into a complete path from external data to analysis and reporting. A ranked list plus skill-gap summary is more actionable than occupation names alone. Grounding the similarity in Salton's Vector Space Model [3] ensures every score is mathematically auditable. The project also makes the source and transformations visible, allowing an evaluator to reproduce and inspect the result.

There are important limitations. O*NET reflects a U.S. framework and may not fully represent Indian hiring, education pathways or local demand [10]. Self-ratings can be wrong due to cognitive biases [13], and skill ratings do not check interests, values, qualifications or work preferences. Similarity depends on feature alignment, scale selection and missing-value handling. Consequently, the tool is best used to explore options and plan learning, not to decide a career automatically.

11. Conclusion

This project applies the Python data science ecosystem to a practical career-guidance problem using an official, downloadable occupational dataset. The pipeline covers data acquisition, validation, cleaning, feature engineering, cosine-similarity ranking, skill-gap reporting and visualisation [1]–[4]. Its main contribution is a reproducible and explainable prototype without a UI. Final numerical findings were generated from the real dataset; the method itself does not prove that a role is right for a particular student.

12. Future Scope

- Merge skills with O*NET interests, knowledge, education and work activities.
- Add Indian labour-market data and regional demand indicators.
- Use structured questions to improve consistency of self-ratings.
- Link gaps to verified free courses and portfolio projects.
- Evaluate recommendations with students and career counsellors.
- Compare ranking methods and test sensitivity to preprocessing choices.

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Dataset attribution

Source: O*NET 31.0 Transferable Skills, O*NET Resource Center, U.S. Department of Labor, Employment and Training Administration. The project transforms the downloaded source into occupational profiles for educational analysis. The source data remain subject to the published O*NET terms and attribution requirements.